

# **Model Assessment (Evaluation/Validation) and Model Selection Part 3**

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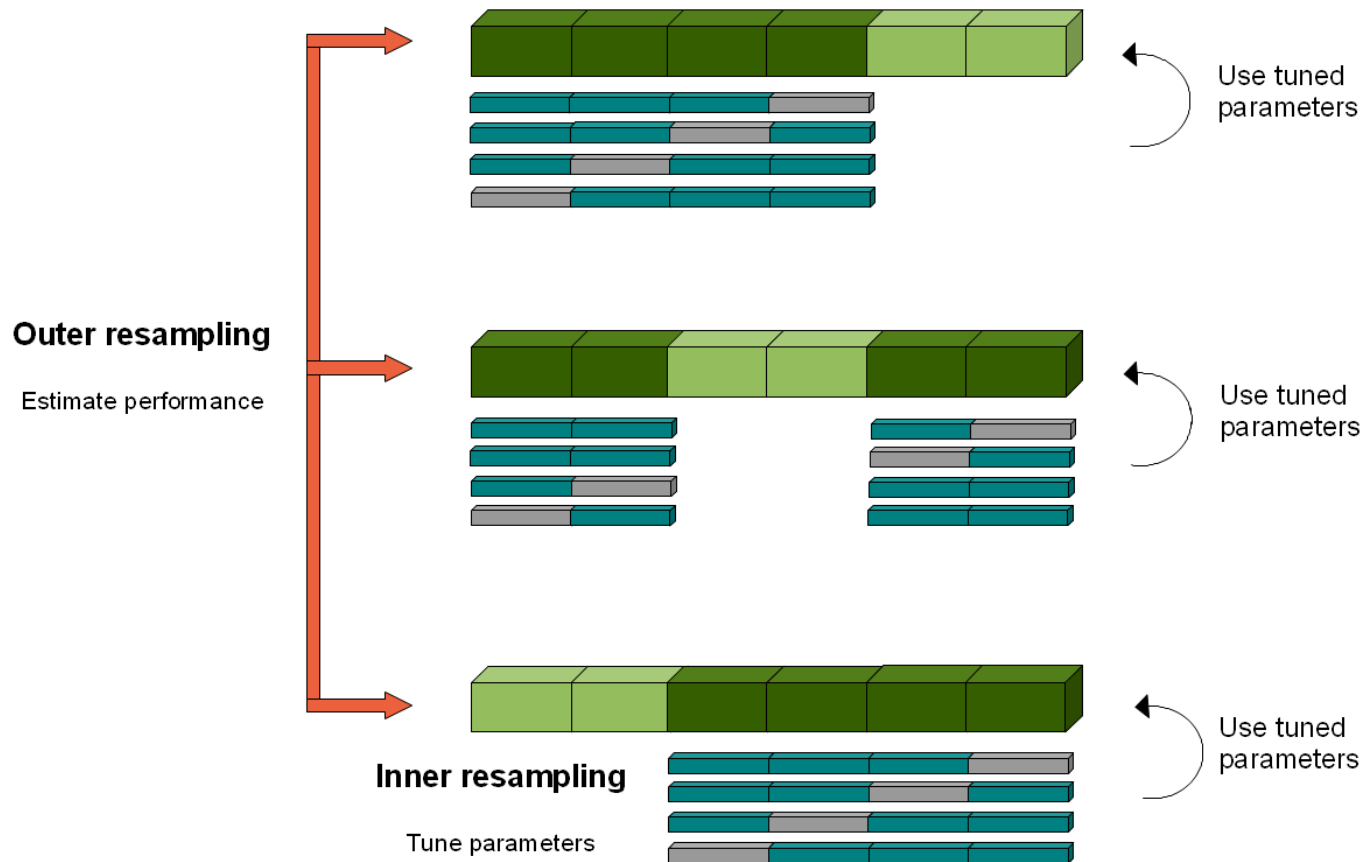
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# Motivation

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- Some further hints on the use of cross-validation
- Typical errors and FAQ on validation
- With some selected slides by **Andrew W. Moore (CMU)**

# Double CV (a different view)



 Training set  
outer resampling

 Test set  
outer resampling

 Training set  
inner resampling

 ~~Test set~~  
inner resampling

aka  
**Validation Set**

# Special cases: *Stopping where?* (FAQ/Frequent **Error**)



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**Avoid** to stop the NN training **fixing an arbitrary number of epochs**

Reasons:

- If it is **small** it may be *too early* (underfitting or just in favor of high eta that need less epochs) or
- if it is **large** it may be *too late* (overfitting or losing time),
- It cannot hold the same value for **all the configurations** and iteration of a K-fold CV, ...
- And what number to use if you **retrain** the model at the end?
- Also selecting the number epochs by *model selection* it is not the best practice: it is a bit better than a fixed number but still *not* accurate, considering the K-fold CV issue and that we have **many other criteria**:
  - See the slides on “Stopping criteria” in the NN-part2 file, where we discussed solutions that works well with a *well regularized system* e.g. looking to the training convergence, i.e. using a threshold for training error decrease or on gradient value, etc. or the *Early Stopping* (ES) if you can/want use a VL set (see also the next slide)

# Special cases: *Early Stopping* (FAQ/Frequent **Error**)



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- **E.S. criteria** should be part of the **model selection** as well (**not** on the TS set, in principle it needs of a **VL set** anytime you train the model).
- It is tricky as it is a strategy that uses part of the data just to *decided when to stop*, without giving a concrete value of a hyperparameter, and it can be complex within CV!
- **Problem:** how to select the best model with ES for the final retraining? (e.g. after a CV you have a different stop point for each validation set)
  - **If** you consider the #epochs a possible hyperpar. (E.g. Mitchell book p.112), you can take the mean #epochs over the folds. **Issue:** varying the data (size) for retraining after model selection could lead to a different (best) stop point! What do you risk? \*
  - So, much BETTER, if you consider the *mean of TR error* at the best point for VL and use it to gain the same level of fitting in the retraining (avoiding undertraining effects).
  - **Else** you need a VL set anytime you are training the model.
  - For instance: CV, ES activated for each validation fold (together with the other hyperpar.). After you get the best hyperparameters values estimation, you retrain the model over the recomposed dataset still using part of the data to find the best for ES (VL set again).
- It is your choice: Using ES or prefer to relay on regularized models with standard stop criteria (e.g. no improvement in training). In any case regularization helps to avoid high overtraining and the need of strict dependences on ES.

# Special cases: *Random init.* (FAQ/Frequent **Error**)



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Random weight initialization (MLP) and model selection

- Different  $\mathbf{W}_{\text{initial}}$   $\rightarrow$  different models
- **Problem:** how to choose one for the final model? (*and you must report the final model choice in the report*)
- You compute the *mean* and *variance* of error/accuracy for the different trials  $\rightarrow$  a random case will be in this range. Hence, the risk is under control unless we get high variance! (but high variance is bad in general - see bias/variance lecture)
- You can take all of them (see "Multiple minima" slide and *ensemble approaches*, see again bias/variance lecture)
- If you chose one of them: it is a *model selection*  $\rightarrow$  you need a VL set
  - the best (i.e. with minimum error on the VL set)
  - the median (to use more "cautious")
  - random (in that case you don't need a VL set, see above)
    - you can also avoid unlucky cases just looking to training error
- More sophisticated approaches are not central for this class.

# Other Typical *errors:* *Sequential selection*



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- **Sequential selection** of the hyperparameters: the order introduce a bias. E.g. eta on 10 units used also for 100 units (!!!)
- After initial trials to determine the right *range/interval* (e.g. by looking to stability of training), the most influential hyperparameters (e.g. looking to variance of results) , etc., the search should be performed **on a grid** with *all the combinations* of the hyper. (of course higher #hyper. and #values → larger table!).
- The grid-search can be repeated to refine the search in a *smaller range* found in the previous grid-search (**nested** grid searches, from coarse to fine grain) so to reduce the effort, but **all** the significant hyper. must be considered together (to account the cross effects).
- But....**Think**, select relevant aspects, don't try with (just) automatic systematic expensive approach.
- Time constraints of the prj should force you to search for basic and fast schema, while still correct /reasonable.

# FAQ on validation/assessment: the final reflection (repetita)



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What to do after model selection and model assessment ?

Can I decide to make a new design on such basis ?

- **Now** you know this is *not* the correct way, it is not really advantageous (unless you will have new data to make a new test set)
  - Repeating this approach with the same data you can achieve any result that you like **fitting this data set** (your model can be perfect on it and not in the future use, you are not estimating the risk).
  - In other words: We shouldn't manually or automatically tune our predictor based on the test error, because we wouldn't be able to do that on future examples.
  - **E.g. Blind test** in competition does not allow you to check and make redesign !
  - But your model assessment (internal test set) is useful to check your **approach!**
- It is well-known the people are tempted to decide on the basis of the results, but this is a “psychological” problem outside of ML methods ;-)
  - At least be aware that you are doing it (and be reasonable) **OR better ....:**
- ❖ If you like to make a good model you have to make a good model selection
- ❖ If you like to estimate the real performance you must make rigorous test estimation.

# Overfitting CV (repetita but with CV)

Intensive use of cross validation can overfit? Yes.

Also does it retain previous issues? An example on **feature selection (FS)**

- Imagine a dataset with 20-30 records and 1000 attributes.
- You try 1000 linear regression models, each one using one of the attributes.
- The best of those 1000 looks good! (validation set)
- But you realize it would have looked good even if the output had been purely random!
- Or [Wrong use] image to approximate a random target finding less then the true error rate 50%(say 3%) **by k-fold CV** (for testing) after the FS by best correlation with targets
- Why? Selection were made on the basis of all the examples
- What can be done about it? You know: “**left out**” **samples before selecting !**
  - Hold out an additional test set before doing any model selection.
  - Check the best model performs well even on the additional test set.

Model selection CV: Use “CV+test” or “double-CV” (see prev. slides: make a new selection for each fold)

Assessment CV [vs wrong use]: FS for each fold: reliable test estimation:

- You will achieve a (correct) average of ~50% for the random task

# Which kind of CV?

|               | Downside                                                             | Upside                                                            |
|---------------|----------------------------------------------------------------------|-------------------------------------------------------------------|
| Test-set      | Variance: unreliable estimate of future performance<br>Lucky/unlucky | Cheap                                                             |
| Leave-one-out | Expensive.<br>Has some weird behavior                                | Doesn't waste data                                                |
| 10-fold       | Wastes 10% of the data.<br>10 times more expensive than test set     | Only wastes 10%. Only 10 times more expensive instead of R times. |
| 3-fold        | Wastier than 10-fold.<br>Expensivier than test set                   | Slightly better than test-set                                     |
| R-fold        | Identical to Leave-one-out                                           |                                                                   |

Where  $R = l$ , i.e. the number of data

# Model Selection: example 1

## Different algorithms/models



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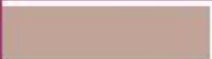
| $i$ | $f_i$ | TRAINERR | 10-FOLD-CV-ERR | VL | Choice |
|-----|-------|----------|----------------|----|--------|
| 1   | $f_1$ |          |                |    |        |
| 2   | $f_2$ |          |                |    |        |
| 3   | $f_3$ |          |                |    |        |
| 4   | $f_4$ |          |                |    |        |
| 5   | $f_5$ |          |                |    |        |
| 6   | $f_6$ |          |                |    |        |

- Whichever model class gave best CV score:  
train It with all the data, and that's the predictive model you'll use.
- Note: the best is chosen looking at the **validation** set result, **NOT** considering (or averaging with) the **TR** error !!!

# Model Selection: example 2

## # hidden units in 1HL-NN

- One hidden layer NN

| Algorithm      | TRAINERR                                                                          | 10-FOLD-CV-ERR                                                                     | VL | Choice                                                                              |
|----------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------|
| 0 hidden units |  |  |    |                                                                                     |
| 1 hidden units |  |   |    |                                                                                     |
| 2 hidden units |  |   |    |  |
| 3 hidden units |  |   |    |                                                                                     |
| 4 hidden units |  |  |    |                                                                                     |
| 5 hidden units |  |  |    |                                                                                     |

- Whichever model class gave best CV score:  
train it with all the data, and that's the predictive model you'll use
- Note; the search can be for 10, 20, 30 units... (or other step size)

# Model Selection: example 3

## K-NN



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- Compute LOOCV error for six different model Classes
- (why LOO?)

| Algorithm | TRAINERR | 10-fold-CV-ERR | VL | Choice |
|-----------|----------|----------------|----|--------|
| $K=1$     |          |                |    |        |
| $K=2$     |          |                |    |        |
| $K=3$     |          |                |    |        |
| $K=4$     |          |                |    | ⊗      |
| $K=5$     |          |                |    |        |
| $K=6$     |          |                |    |        |

# Model Selection: example 3

## K-NN: details



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- Example: Choosing “k” for a k-nearest-neighbor regression.
- Step 1: Compute LOOCV error for six different classes:

| Algorithm | Training Error |
|-----------|----------------|
| K=1       |                |
| K=2       |                |
| K=3       |                |
| K=4       |                |
| K=5       |                |
| K=6       |                |

Why did we use 10-fold-CV for neural nets and LOOCV for k-nearest neighbor?

And why stop at K=6

Are we guaranteed that a local optimum of K vs LOOCV will be the global optimum?

What should we do if we are depressed at the expense of doing LOOCV for K= 1 through 1000?

The reason is Computational. For k-NN (and all other nonparametric methods) LOOCV happens to be as cheap as regular predictions.

No good reason, except it looked like things were getting worse as K was increasing

Sadly, no. And in fact, the relationship can be very bumpy.



Idea One: K=1, K=2, K=4, K=8, K=16, K=32, K=64 ... K=1024

Idea Two: Hillclimbing from an initial guess at K

# Conclusions on practical issues



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In practice:

- Simple Hold Out is widely applied  
Sometimes the test set is already given separately or “taken” by previous applications
  - See Warnings for few data
  - Check your TR and VL error rates
- 5 or 10-fold CV are recommended as a good compromise (among bias and variance of estimation) with respect to LOO (Hastie et al. and the references therein)
- We continue with an analytic approach to approximate the estimation step
- Interesting insights for model selection/comparison (relative size of error is what matters) instead of pure empirical evaluation

# Next Lecture

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Let us now go back to the analytical estimation of the expected error (validation or test error)

**SRM: Structural Risk Minimization & VC-dimension**

*Please, in the while have a look back to the lect. 4 on SLT  
(in the introduction of the course)*

# ML Course structure

## Where we go



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